

Kairix AI

AI-Powered Career Coaching System

Vision AI · Speech AI · RAG · Automation

Muhammad Tahir

AI Product Manager — Cohort 28 · May 2026



Vision AI

Google Cloud Vision API



Speech AI

Web Speech API + TTS



RAG System

LangChain + ChromaDB



Automation

n8n → Gmail + Sheets

Problem & Solution

THE PROBLEM

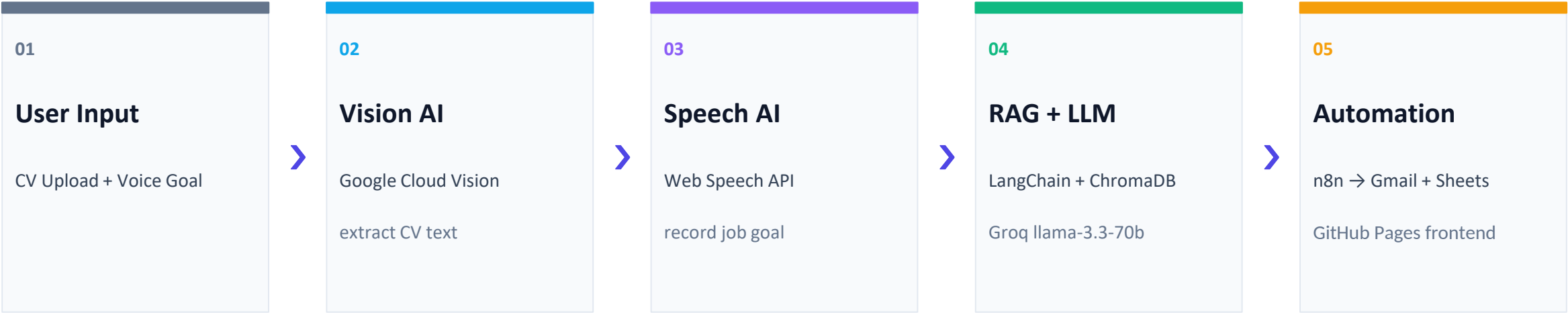
- Job seekers get rejected without knowing why
- Generic CV advice doesn't match real job market requirements
- Career coaching is expensive and slow — feedback takes days
- No affordable tool gives instant, role-specific, scored feedback



THE SOLUTION — KAIRIX AI

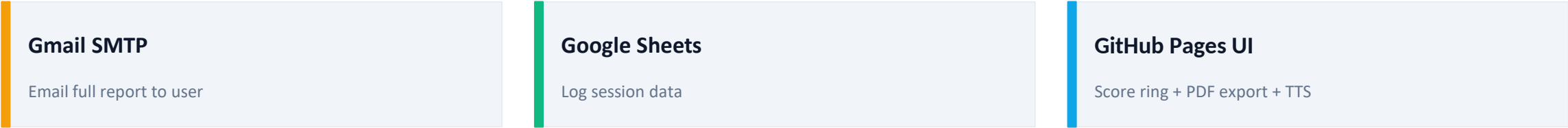
- Upload CV as JPG, PNG, PDF, or DOCX
- Speak your target role via microphone
- Receive a scored coaching report in under 10 seconds
- Full report emailed automatically via n8n + Gmail
- Session logged to Google Sheets for tracking

System Architecture



End-to-end pipeline: user input → AI processing → intelligent output → real-world automation

n8n Webhook splits into:



Four AI Technologies — One Unified System



Vision AI

Google Cloud Vision API

- Analyses uploaded CV images using OCR
- Extracts all text from JPG, PNG, WEBP files
- pdfplumber handles PDF text extraction
- python-docx handles DOCX files



Speech AI

Web Speech API + Web Speech Synthesis

- SpeechRecognition converts voice to text in real time
- User speaks their target job role — no typing needed
- Web Speech Synthesis reads back the full report aloud
- Works natively in Chrome and Edge — no library needed



RAG System

LangChain + ChromaDB + Sentence Transformers

- Custom dataset: 2026 job market data & salary benchmarks
- all-MiniLM-L6-v2 embeds documents into ChromaDB
- Relevant context retrieved at query time via LangChain
- Injected into LLM prompt for role-specific accuracy



Automation

n8n Webhook → Gmail SMTP + Google Sheets

- n8n webhook triggered after every analysis
- Full coaching report emailed to user automatically
- Session logged: timestamp, email, job goal, score
- Zero manual intervention — fully event-driven

Tools & Technologies

Frontend	HTML5, CSS3, Vanilla JavaScript	Embeddings	Sentence Transformers — all-MiniLM-L6-v2
Voice Input	Web Speech API — SpeechRecognition	Vector Store	ChromaDB
Text-to-Speech	Web Speech Synthesis API	RAG Framework	LangChain
Backend	FastAPI — Python 3.10+	AI Model	Groq — llama-3.3-70b-versatile
CV OCR (Images)	Google Cloud Vision API	Automation	n8n → Gmail SMTP + Google Sheets
CV Extraction (PDF)	pdfplumber	Frontend Hosting	GitHub Pages
CV Extraction (DOCX)	python-docx	Backend Hosting	Hugging Face Spaces

Live Demo & Repository

LIVE DEMO

muhammادتahir-1-9-4.github.io/kairix-ai

Fully deployed — no login required · works in Chrome / Edge

Analysis completes in under 10 seconds

GITHUB REPOSITORY

github.com/MuhammadTahir-1-9-4/kairix-ai

Full source code · FastAPI backend · HTML/CSS/JS frontend

README with setup, architecture, and API reference

DEMO WALKTHROUGH

- 1 Open the live demo in Chrome
- 2 Upload a CV (JPG / PDF / DOCX)
- 3 Click microphone → speak target role
- 4 Click 'Analyze My Career'
- 5 View score ring, strengths, skill gaps
- 6 Click 'Hear Your Feedback' — TTS plays
- 7 Download PDF report
- 8 Check email — report delivered via n8n

What Kairix AI Delivers

AI OUTPUT — PER ANALYSIS

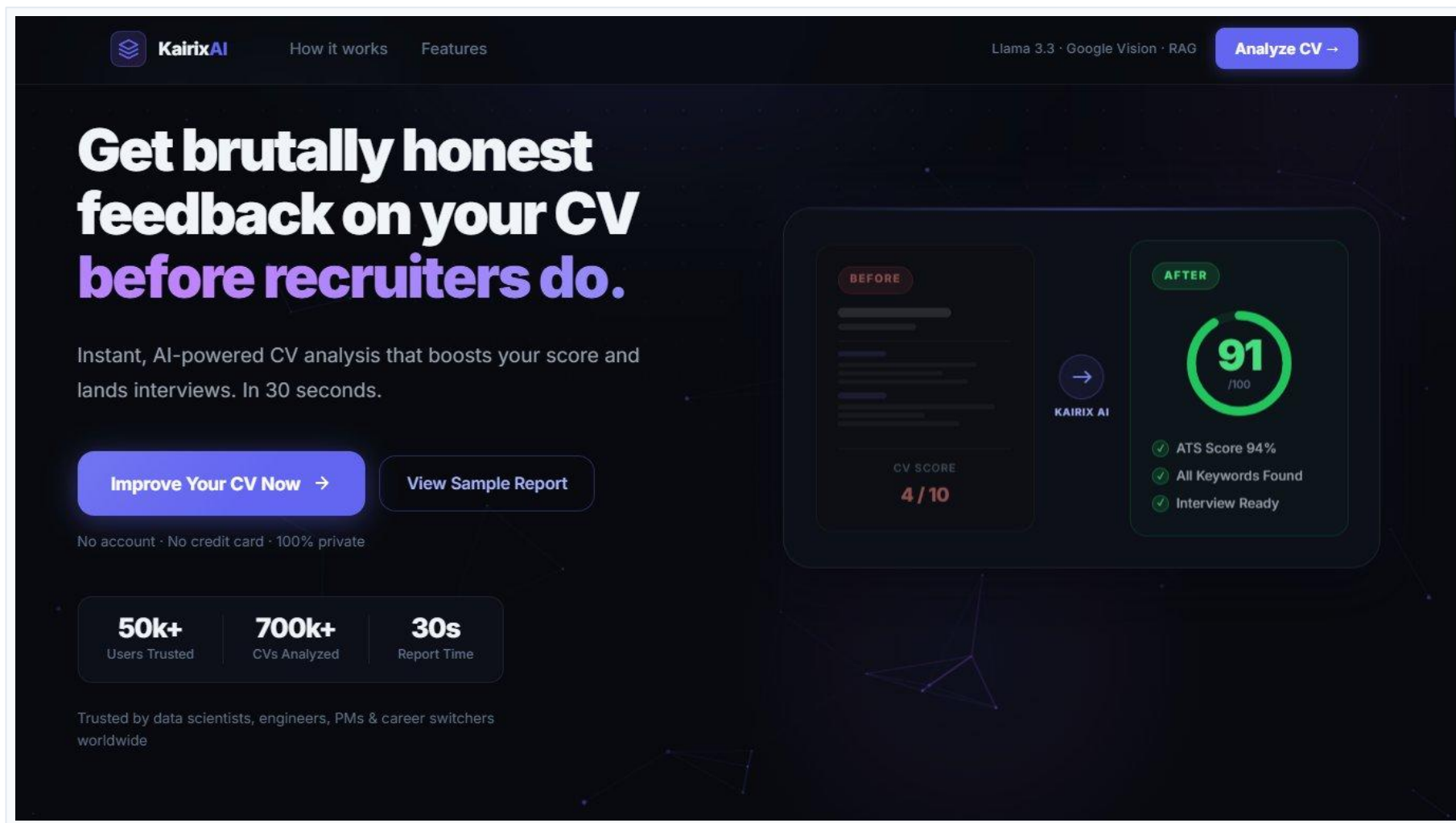
- CV Score out of 100 — visual ring indicator
- Score explanation — why you received that score
- Strengths — what your CV does well
- Skill Gaps — what is missing for the target role
- CV Improvements — specific line-by-line suggestions
- Next Steps — certifications, projects, and actions
- Motivational message personalised to the user
- Text-to-speech playback of the full report
- PDF export of the complete coaching report

AUTOMATION — EVERY SESSION

- n8n webhook triggered automatically on completion
- Full report emailed via Gmail SMTP — no user action needed
- Session logged to Google Sheets: timestamp, email, score
- Fully event-driven — zero manual intervention

EVALUATION CRITERIA — COVERED

- ✓ **Innovation** — novel AI career coaching product
- ✓ **Technical integration** — 4 AI systems unified
- ✓ **End-to-end pipeline** — fully functional in production
- ✓ **Real-world problem** — career services domain



Upload Form — 4-Step Cards

ANALYZE

Get your **Career Intelligence Report**

Free · Private · No account required · Under 30 seconds

1 Upload CV

2 Your Goal

3 Match JD

4 Generate

01 Upload your CV

Accepted: JPG, PNG, PDF, DOCX · Max 10 MB



Drop your CV here, or [browse](#)

PDF · DOCX · JPG · PNG · WEBP

02 Describe your target role

Speak your goal out loud or type it below

 Click to record your goal

Works best in Chrome

"I want to become a senior data analyst at a fintech company in London"

OR TYPE YOUR GOAL

e.g. I want to transition into product management at a B2B SaaS company, ideally from my current software engineering role.

03 Paste a job description

OPTIONAL · UNLOCKS ATS MATCH

Get a keyword match % and ATS analysis against a real job posting

Paste the full job description here — requirements, responsibilities, preferred qualifications...

The AI will simulate an ATS scan and tell you exactly which keywords you're missing.

04 Email your report

OPTIONAL

We'll send the full coaching report to your inbox

you@example.com

No spam. No account required. Unsubscribe anytime.

Generate My Career Report

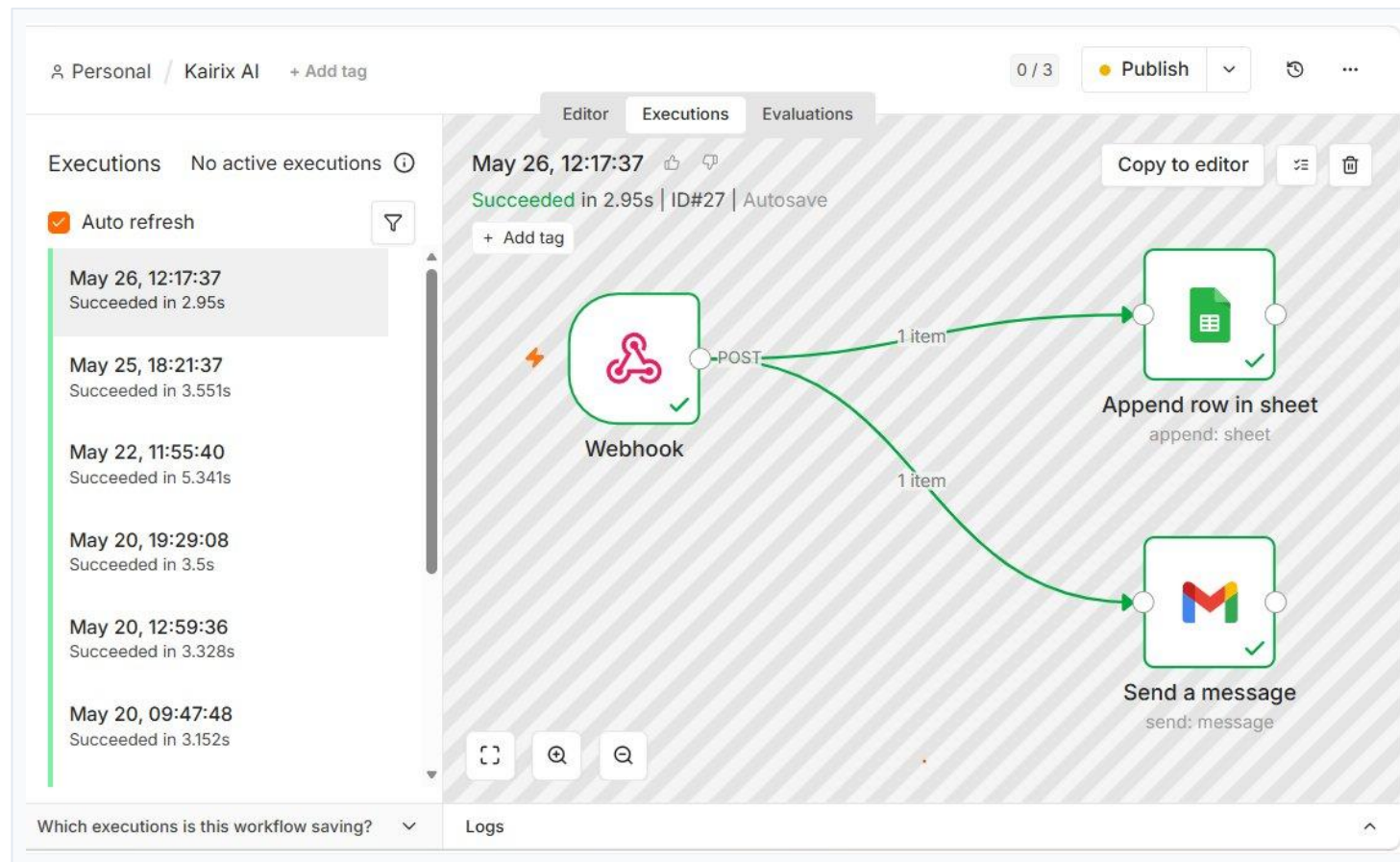
Upload your CV and describe your goal to continue

Results Screen — Career Intelligence Report

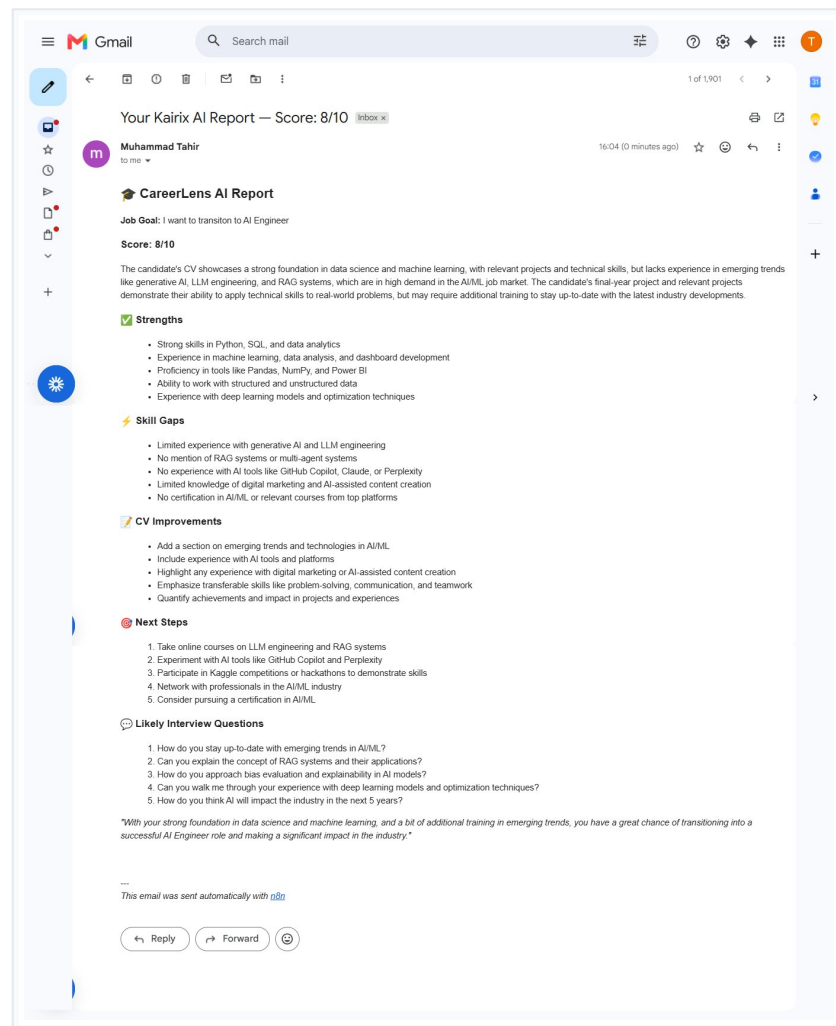
Live analysis output · score ring, JD match, strengths, skill gaps, CV improvements



Production workflow on n8n · Webhook trigger splits into Google Sheets logging and Gmail report delivery

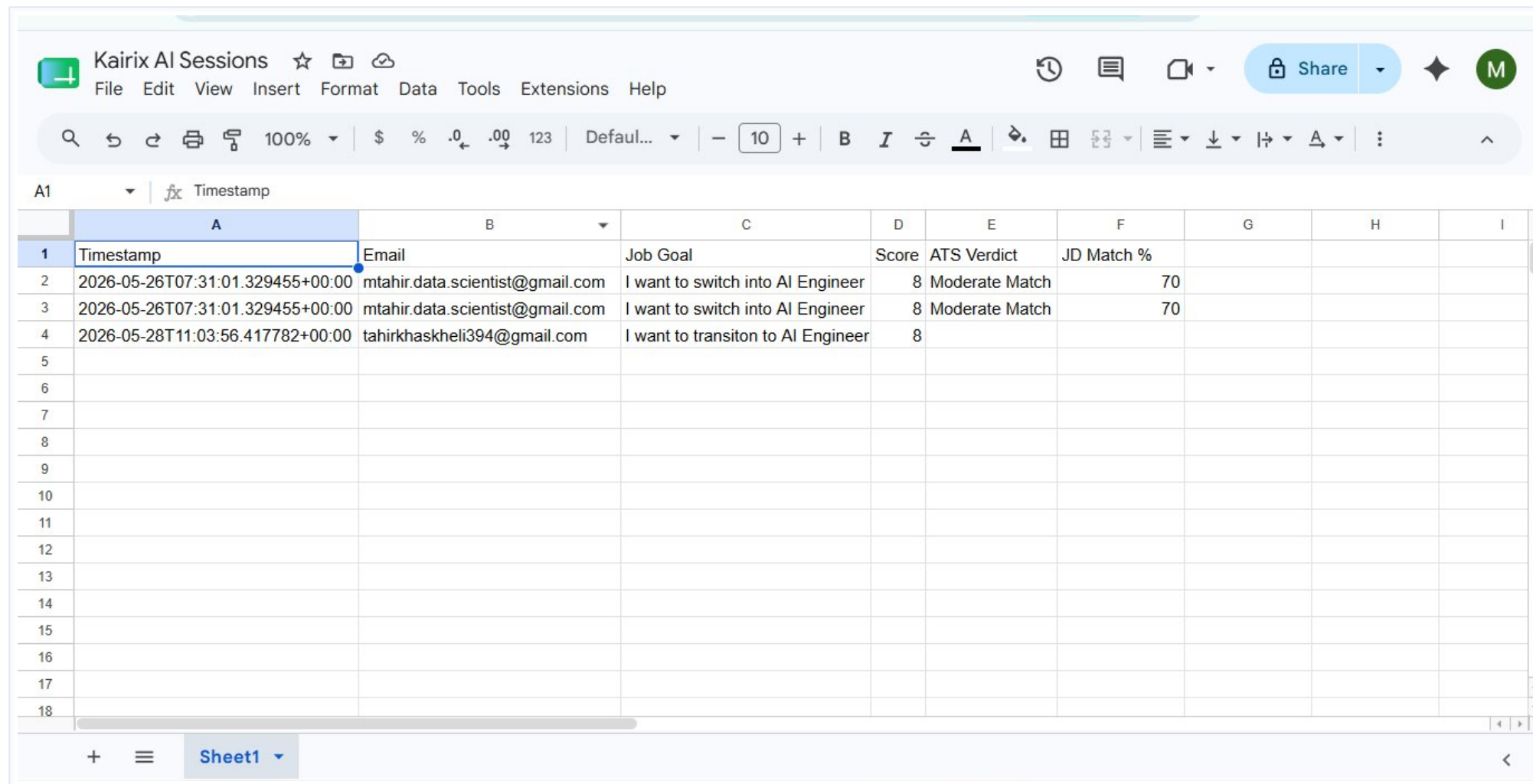


Coaching report delivered automatically by n8n · score, strengths, skill gaps, next steps, interview Qs



Google Sheets Session Log

Every analysis appended in real time by n8n · timestamp, email, job goal, score, ATS verdict, JD match %



The screenshot shows a Google Sheets interface with the title 'Kairix AI Sessions'. The spreadsheet contains a table with the following data:

	A	B	C	D	E	F	G	H	I
1	Timestamp	Email	Job Goal	Score	ATS Verdict	JD Match %			
2	2026-05-26T07:31:01.329455+00:00	mtahir.data.scientist@gmail.com	I want to switch into AI Engineer	8	Moderate Match	70			
3	2026-05-26T07:31:01.329455+00:00	mtahir.data.scientist@gmail.com	I want to switch into AI Engineer	8	Moderate Match	70			
4	2026-05-28T11:03:56.417782+00:00	tahirkhaskheli394@gmail.com	I want to transition to AI Engineer	8					
5									
6									
7									
8									
9									
10									
11									
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14									
15									
16									
17									
18									

GitHub Repository

**kairix-ai**

Public



Pin



Watch 0



Fork 0



Star 0



main



1 Branch



0 Tags



Go to file



Code

**MuhammadTahir-1-9-4**

Point frontend to HF Space backend, fix status mess...



f1f1cc1 · 15 hours ago



23 Commits



.github/workflows

Add GitHub Pages deployment workflow

16 hours ago



backend

Point frontend to HF Space backend, fix status messa...

15 hours ago



frontend

Point frontend to HF Space backend, fix status messa...

15 hours ago



.gitignore

Add portfolio presentation PDF and update gitignore

17 hours ago



Kairix_AI_Portfolio2.pdf

Add portfolio presentation PDF and update gitignore

17 hours ago



README.md

Update README: repo URL updated to kairix-ai

yesterday



netlify.toml

Add netlify.toml to set publish directory

2 weeks ago



README



About

kairix-ai.vercel.app

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Deployments 7

github-pages 40 minutes ago **Production** 15 hours ago